

2

Swarm Level: Small-World Multi-Agent Network

WATTS-STROGATZ TOPOLOGY • O(N) WHITEBOARD

🌐 swarm-topology-builder

Constructs clustered domain teams linked by sparse shortcut paths, preventing O(N²) communication broadcast storms.

✉️ swarm-liaison-router

Designated boundary agents filter, authenticate, and bridge cross-cluster state transitions with ACL enforcement.

⌚ swarm-hop-limiter

Attaches monotonic TTL headers (MaxHops: 4) to RPC payloads to eliminate infinite inter-agent delegation loops.

📄 swarm-shared-whiteboard

Centralized O(N) shared blackboard with DynamoDB CAS mutex leases; guarantees no two agents duplicate the same task.

🗣️ swarm-consensus-arbiter

Coordinates multi-agent debate, cross-verifications, and cryptographic vote tallying for final state consensus.

🛡️ swarm-circuit-breaker

Global supervisor detecting runaway ping-pong loops; quarantines stuck nodes without halting the enterprise swarm.

▼ COMPOSITION: Swarm nodes instantiate isolated Level 1 Agent Kernels

1

Agent Kernel Level: The Individual OS Harness

MODEL PROPOSES • KERNEL DECIDES • RING 0 SAFETY

🧠 EXECUTIVE CONTROL SUITE (Prefrontal Cortex)

DETERMINISTIC GATES

🔑 kernel-propose-decide

Ring 0 syscall interceptor. The model proposes; the kernel validates schemas and grants execution.

📖 kernel-runbook-engine

Executes SKILL.md runbooks with automated <30ms contract tests; prevents arbitrary workflow drift.

⚡ kernel-executive-supervisor

Watchdog tracking sliding-window tool repetition; kills loops within 3 iterations and trips breakers.

💾 MULTI-STORE MEMORY SUITE (Hippocampal/Cortical)

FAITHFUL BY DESIGN

📁 kernel-working-memory-pager

Context window RAM manager. Pages relevant history, limits token bloat, and snapshots state.

📅 kernel-episodic-ledger

Append-only S3 time-series record of turns, enabling replayable deterministic debugging.

🔗 kernel-grounding-certifier

Hybrid Vector + Graph certifier; blocks hallucinated quotes lacking signed S3 cryptographic receipts.

▼ COMPOSITION: Kernel delegates physical operations to Level 0 AWS Primitives

0

Base Level: AWS Cloud Primitives & Simulator Bindings

HARDWARE & INFRASTRUCTURE ROOTS • WORM COMPLIANCE

🧠 aws-bedrock

Foundation model inference & structured schema proposals via Claude / Llama.

🛡️ aws-agentcore

Ring 0 policy enforcement, Cedar guardrails, and IAM permission boundaries.

💾 aws-s3-vault

Immutable WORM compliance storage with SHA-256 grounding receipts.

📄 aws-database

DynamoDB distributed mutex locks, OpenSearch vectors, and Neptune graphs.

⚡ aws-lambda

Stateless Firecracker microVM tool execution & isolated code evaluation.

🕒 aws-stepfunctions

Deterministic HFSM state machines executing invariant-governed runbooks.

👤 aws-client-api

EventBridge asynchronous event bus and API Gateway endpoints for operator ingress, telemetry, and webhook distribution.